# JavaScript Coding Problems for Practice

## 1. Variables & Data Types

1. Create a program that swaps two numbers without using a third variable.
2. Write a program to check whether a given value is a number, string, boolean, null, or undefined.
3. Convert temperature from Celsius to Fahrenheit using variables.
4. Create a simple calculator using variables and arithmetic operators.
5. Write a program that takes a user's birth year and calculates age.

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## 2. Strings

1. Reverse a string without using built-in reverse methods.
2. Count the number of vowels in a string.
3. Check whether a string is a palindrome.
4. Capitalize the first letter of every word in a sentence.
5. Find the longest word in a sentence.

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## 3. Numbers & Math

1. Generate a random number between 1 and 100.
2. Check whether a number is prime.
3. Find factorial of a number using loops.
4. Find Fibonacci series up to `n` numbers.
5. Check whether a number is Armstrong number.

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## 4. Arrays

1. Find the largest and smallest number in an array.
2. Remove duplicate elements from an array.
3. Sort an array without using built-in `sort()`.
4. Find second largest number in an array.

5. Merge two arrays and remove duplicates.

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## 5. Objects

1. Create an object for a student and display all properties dynamically.
2. Count number of keys in an object.
3. Merge two objects into one.
4. Convert an object into an array of keys and values.
5. Create a shopping cart object and calculate total bill amount.

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## 6. Loops

1. Print star pyramid patterns.
2. Print multiplication tables from 1 to 10.
3. Find sum of all even numbers between 1 and 100.
4. Print all prime numbers between 1 and 100.
5. Create a number guessing game using loops.

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## 7. Functions

1. Create a function that returns the greatest of three numbers.
2. Write a function to check whether a number is palindrome.
3. Create a reusable function for currency conversion.
4. Write a function that accepts an array and returns only even numbers.
5. Create a calculator using functions for add, subtract, multiply, and divide.

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## 8. Scope & Hoisting

1. Predict output of variable hoisting examples.
2. Create examples showing block scope using `let`.
3. Write a program demonstrating closure behavior.
4. Create nested functions and access outer variables.
5. Debug a program with incorrect variable scoping.

## 9. Callback Functions & `setTimeout`

1. Create a delayed greeting message using `setTimeout`.
2. Build a countdown timer.
3. Create a callback-based calculator.
4. Simulate food ordering system using callbacks.
5. Execute functions sequentially using callbacks.

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## 10. Arrow Functions

1. Convert normal functions into arrow functions.
2. Create one-line arrow functions with implicit return.
3. Use arrow functions with `map()`.
4. Build an even/odd checker using arrow functions.
5. Create a student grade calculator using arrow functions.

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## 11. `map()`, `filter()`, `forEach()`

1. Double all numbers in an array using `map()`.
2. Filter all students scoring above 80 marks.
3. Print all array values using `forEach()`.
4. Convert array of names into uppercase.
5. Extract only even numbers using `filter()`.

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## 12. Primitive vs Reference Types

1. Create examples showing primitive copying behavior.
2. Create examples showing object reference behavior.
3. Clone an object without affecting original object.
4. Compare arrays using reference equality.
5. Demonstrate shallow copy using spread operator.

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## 13. Stack & Heap Memory

1. Create examples showing stack memory behavior.
2. Create examples showing heap memory references.
3. Demonstrate object mutation through references.
4. Predict outputs of memory-related programs.
5. Create diagrams explaining memory allocation.

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## 14. Call Stack

1. Trace execution order of nested function calls.
2. Create recursive factorial function.
3. Build recursive Fibonacci function.
4. Simulate stack overflow with recursion.
5. Draw call stack flow for nested functions.

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## 15. JSON Problems

1. Convert JSON string into object.
2. Convert object into JSON string.
3. Parse API-like JSON data and display values.
4. Create a JSON array of employee records.
5. Filter JSON data based on conditions.

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# Mini Project-Level Coding Challenges

## Beginner Projects

1. Student Grade Calculator
2. ATM Withdrawal Simulator
3. To-Do List using Arrays
4. Password Generator
5. Quiz Application

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## Logic Building Challenges

1. Find missing number in array.
2. Check whether two strings are anagrams.
3. Rotate array by `k` positions.
4. Find duplicate elements in array.
5. Flatten nested arrays manually.

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# Debugging Challenges

1. Fix an infinite loop issue.
2. Identify why `undefined` is returned from a function.
3. Debug incorrect array sorting behavior.
4. Resolve scope-related variable issues.
5. Fix callback execution order problem.